Ejemplo de Livescript

Sesion 1:

Comandos de inicio:

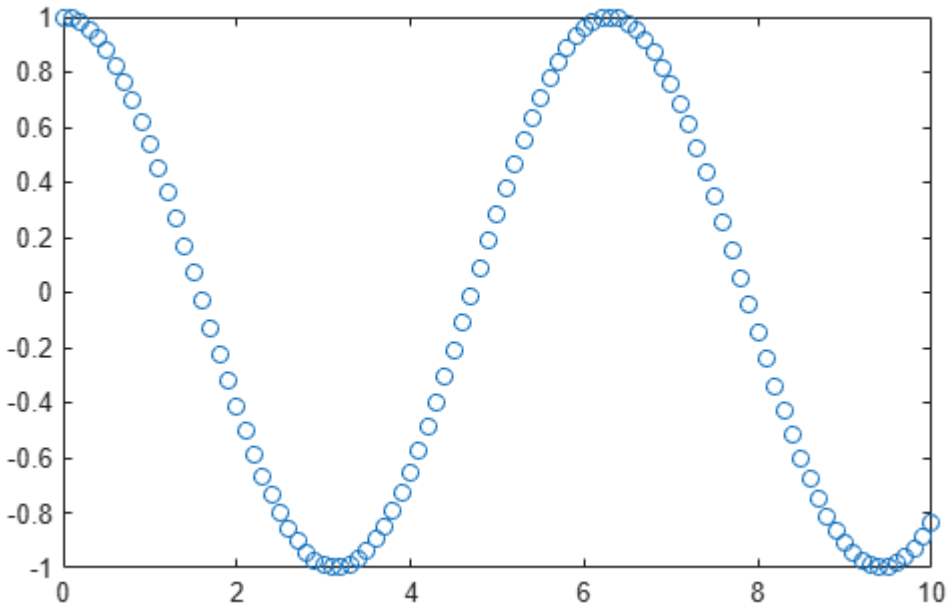
```
clc; clear; close all  % Importante ponerlo siempre
```

Vectores:

```
vec1 = 0:0.1:10  %Vector del 1 al 10 con numeros incrementandose 0.1
```

```
vec1 = 1×101
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y = cos(vec1);
plot(vec1, y, 'o')
```



Ejemplo funcion 3D:

```
x = 0:0.1:10
```

```
x = 1×101
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

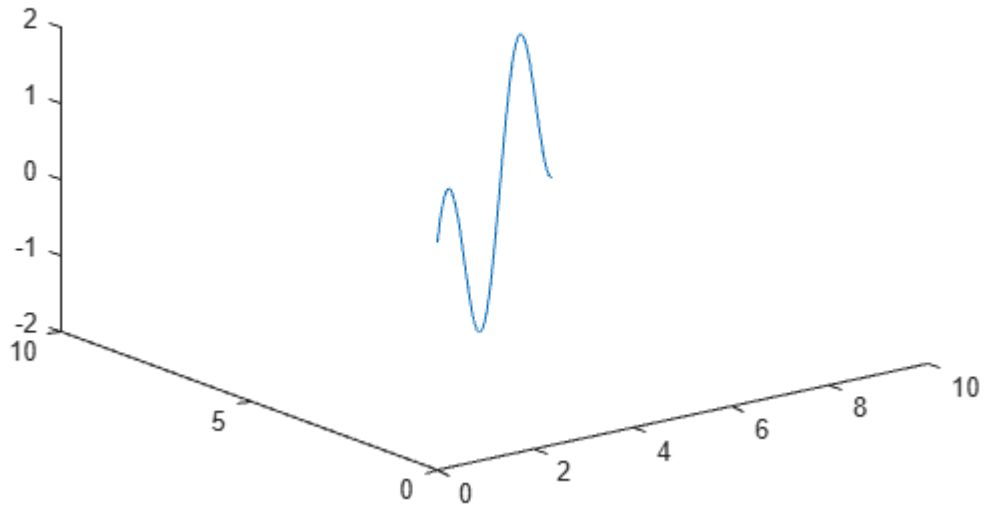
```
y = 0:0.1:10
```

```
y = 1×101
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
z = cos(x) + sin(y)
```

```
z = 1×101
    1.0000    1.0948    1.1787    1.2509    1.3105    1.3570    1.3900    1.4091 ···
```

```
plot3(x,y,z)
```



Ejemplo hold on:

```
t = 0:0.1:20
```

```
t = 1×201
         0    0.1000    0.2000    0.3000    0.4000    0.5000    0.6000    0.7000 ···
```

```
y1 = cos(t)
```

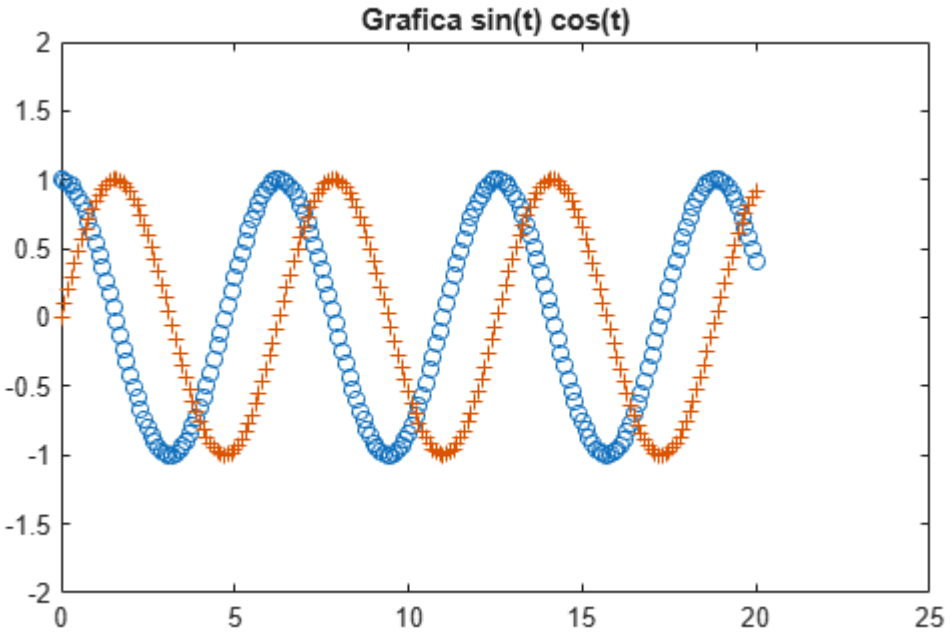
```
y1 = 1×201
    1.0000    0.9950    0.9801    0.9553    0.9211    0.8776    0.8253    0.7648 ···
```

```
y2 = sin(t)
```

```
y2 = 1×201
         0    0.0998    0.1987    0.2955    0.3894    0.4794    0.5646    0.6442 ···
```

```
plot(t,y1,'o')
title("Grafica sin(t) cos(t)")
axis([0 25 -2 2])
hold on % Representa las dos funciones juntas

plot(t,y2,'+')
hold off
```

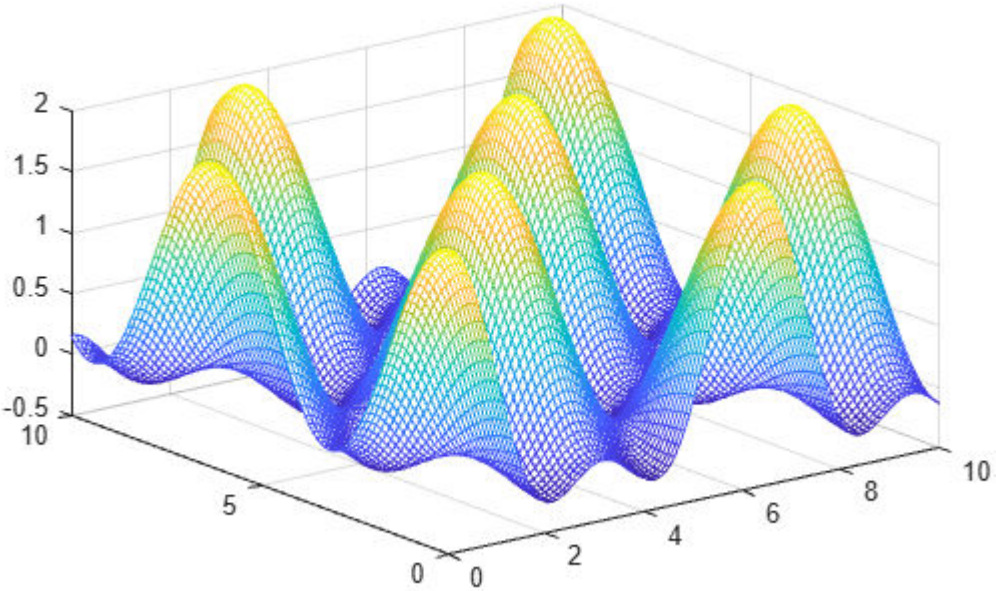


Ejemplo maya (meshgrid):

```
%Valores x e y
x = 0:0.1:10;
y = 0:0.1:10;
% Creamos la maya
[X, Y] = meshgrid(x,y);

Z = cos(X).*cos(x)+cos(y).*cos(Y);

mesh(X,Y,Z) %Representacion de la funcion
```



```
surf(X,Y,Z) %Rellena los huecos de la maya
colormap NEBULA % Colorear la funcion
```

